Recording a Voice for Conversion

Session guide. The recording is the ceiling on everything downstream — no amount of processing recovers detail the microphone never captured.

Why this sheet exists

The voice model on this project reproduced the singer accurately. It was measured against his own recordings and tracked them within a few dB in every frequency band. The problem was that the recordings themselves were far darker than a studio vocal:

Frequency band	Studio vocal	Phone take	Shortfall
2–4 kHz — presence, the "front" of a voice	-9.9 dB	-26.1 dB	**16 dB**
8–11 kHz — air, the sheen on top	-26.4 dB	-55.0 dB	**28 dB**

The model learned that dullness faithfully, because that is what it was given. Mix processing later recovered the presence band, but the air above 8 kHz had to be synthesised — there was nothing real up there to lift. A better recording fixes at the source what everything else can only imitate.

The room: dry, not resonant

This is the one place instinct points the wrong way. A stairwell, a bathroom or a church makes singing feel and sound wonderful in the moment — and each is close to the worst possible choice for this purpose.

- Reverb is recorded **into** the voice. The model cannot tell a room from a throat, so it learns the room as part of the timbre and then carries it into every song afterwards.
- It cannot be removed later. De-reverb processing is lossy and audibly damages the voice.
- The mix adds its own reverb at the end. A reverberant source gets it twice and turns to mush.
- Small hard rooms — bathroom, stairwell — are the worst of all. Their strong early reflections comb-filter the direct sound, which eats exactly the high frequencies this recording is trying to gain.

Record here	Not here
A closet, standing among hanging clothes	Bathroom or shower room
A bedroom with a rug, curtains and a bed	Stairwell
Under a thick blanket draped over you and the mic	Church or hall
A parked car, engine off	Any large empty room

The clap test. Stand where you plan to sing and clap once, hard. If you hear any ring or tail after the clap, the room is too live — add soft material or move. A good space sounds slightly dead and unflattering.

The instinct is not worthless: those spaces are usually quiet, and they make people sing with more confidence. Warm up in the stairwell if it helps. Then record somewhere dull.

The microphone

- **Any real microphone beats a phone held at arm's length.** A large-diaphragm cardioid condenser is ideal; a handheld dynamic (SM58 type) is a fine second choice and more forgiving in a live room.
- **Distance: 15–20 cm**, and keep it identical across every take. Closer gets boomy, further picks up the room and loses top end.
- **Point your mouth straight at the capsule.** High frequencies are directional and fall away fast off-axis. Singing past the side of a mic is a common cause of a dull recording, and likely part of the 28 dB shortfall measured above.
- **Use a pop filter**, or stretch a sock over the mic, and aim slightly past the capsule so plosives do not thump it.
- **Same mic, same distance, same room for every take.** Mixed conditions average out into a timbre that is nobody's.

Recorder settings

Setting	Value	Why
Sample rate	48 kHz	One earlier take arrived at 16 kHz, which has no content at all above 8 kHz. It could not be used and was dropped.
Bit depth / format	24-bit WAV, or the app's lossless option	Compression artefacts get learned like anything else.
Noise suppression / voice isolation	OFF	It removes breath and high detail, which is precisely what is wanted.
EQ, compression, normalise, presets	OFF	Hand over the raw file.
Level	Peaks around -6 to -12 dBFS	Never let it clip. Clipping is unrecoverable.

Using the guide track

Wear headphones. This is not optional. If the guide tone plays out of a speaker it is recorded along with the voice and becomes part of what the model learns.

- The track walks up one semitone at a time, so you are never asked to jump to a note out of nowhere. On a previous session a singer followed a guide that jumped straight to the target and sang the whole run **a perfect octave too low** without noticing — it felt right, because the note names matched. Walking up in steps prevents that.
- **Three passes through the whole range.** Sing "ah" on the first pass, "ee" on the second, "oo" on the third.
- Hold each note for as long as the tone lasts, roughly three and a half seconds. Land on the pitch rather than sliding up into it.
- **A short high pip marks every C**, so you can tell where you are without counting.
- **When it climbs past your range, or drops below it, just stay silent.** Let the track run. Silence costs nothing; a strained squeak teaches the model a voice you do not have.
- One continuous recording for the whole track is fine. Do not stop and start between notes.

Sing in chest or mixed voice, not breath

Breathy falsetto has almost no resonance, so the model learns nothing from it and the top of the range still comes out hollow. **A strained, cracking, honest note is worth more than a pretty airy one.** This is genuinely counter-intuitive and worth saying out loud before the take starts.

Where your range is likely to end

The track runs G2 to C5, wide enough for most male voices and for the songs in hand. Expect to drop out at one end or both; that is the point of recording the whole thing once.

Voice type	Typical range	Voice type	Typical range
Bass	E2 – E4	Alto	F3 – F5
Baritone	G2 – G4	Mezzo	A3 – A5
Tenor	C3 – C5	Soprano	C4 – C6

Check the first take before recording a second

Send the very first file over before doing anything else. An octave error, a wrong sample rate or a live room takes seconds to spot and invalidates everything recorded after it. On this project that check would have saved an entire session.

What to send back

- The raw files, unedited, in whatever format they were recorded.
- A one-line note: which microphone, which room, roughly how far away.
- Ten to fifteen minutes of actual singing is plenty. Three passes of the guide gives about ten seconds per semitone, which is the target.